MindSync AI
10-15 Minute Viva Presentation & Live Demo Guide

AI Mental Wellness Companion for IT Professionals

Master Presentation Script, Slide Deck & Live Demo Workflow

Generated: August 15, 2026

Target Presentation Duration: 10 - 15 Minutes

Includes: 11 Slide Deck Scripts, Click-by-Click Live Demo Instructions, Technical Cheatsheet & Q&A Defense

SECTION 1: 10-15 MINUTE PRESENTATION TIMELINE

Below is the exact time allocation recommended for a 15-minute presentation committee slot. Adjust timing slightly if given 10 minutes.

Time Block	Segment	Focus Area	Slide(s)
0:00 - 1:00 (1 min)	Introduction	Title, Project Overview & Key Value Proposition	Slide 1
1:00 - 2:30 (1.5 min)	Problem Statement	IT Occupational Burnout Crisis & Industry Need	Slide 2
2:30 - 4:00 (1.5 min)	System Architecture	Three-Tier Hybrid AI Design (ML + GenAI + Rules)	Slide 3
4:00 - 5:30 (1.5 min)	ML & Explainable AI	Random Forest Model, RobustScaler & SHAP TreeExplainer	Slides 4 - 5
5:30 - 7:00 (1.5 min)	Generative AI & Rules	Groq LLaMA 3.3-70B, Prompt Security & Rule Engine	Slide 6
7:00 - 11:30 (4.5 min)	LIVE DEMO	Step-by-Step Application Walkthrough (Flutter App)	Slide 7 - 8
11:30 - 13:00 (1.5 min)	Security & DevOps	Firebase, JWT Auth, Docker & Render Cloud Deployment	Slide 9
13:00 - 15:00 (2.0 min)	Conclusion & Q&A	Key Takeaways, Testing Metrics (93% Acc) & Defense	Slides 10 - 11

SECTION 2: SLIDE-BY-SLIDE PRESENTATION SCRIPT

Slide 1: Title & Introduction

Timing: 0:00 - 1:00 min

Speaker Script: "Good morning respected examiners and members of the panel. Today, I am presenting MindSync AI -- an AI-driven mental wellness companion engineered specifically for software engineers and IT professionals."

Slide Visual Highlights:

* Project Name: MindSync AI

* Target Audience: Software developers, DevOps engineers, and IT professionals

* Key Feature Matrix: ML Burnout Prediction, Explainable AI (SHAP), Groq LLM Coaching, Hybrid Recommendations, Telemetry Dashboards

* Tech Stack: Flutter (Frontend), FastAPI Python (Backend), Random Forest + SHAP (ML), Groq LLaMA 3.3-70B (GenAI), Firebase (Cloud DB/Auth)

Slide 2: Problem Statement & IT Burnout Crisis

Timing: 1:00 - 2:30 min

Speaker Script: "According to industry research, over 83% of software developers suffer from occupational burnout. Long coding sessions, tight sprint deadlines, late-night debugging, and constant screen time lead to severe cognitive fatigue and mental exhaustion."

Slide Visual Highlights:

* Existing Wellness Apps: Generic (meditation-only), ignore developer-specific stress triggers, lack predictive capabilities

* MindSync AI Solution: Data-driven predictive model tracking developer metrics (sleep, stress, working hours, consecutive days, exercise, hydration)

* Core Innovation: Combines predictive Machine Learning with Explainable AI so developers know WHY they are at risk

Slide 3: Three-Tier Hybrid AI Architecture

Timing: 2:30 - 4:00 min

Speaker Script: "To guarantee system reliability and high-quality user assistance, MindSync AI utilizes a robust Three-Tier Hybrid AI Architecture."

Slide Visual Highlights:

* Tier 1 - Classical Machine Learning: Random Forest model predicts burnout risk level (Low, Medium, High) with 93% accuracy

* Tier 2 - Generative AI: Groq LLM (LLaMA 3.3-70B) generates context-aware, creative habit recommendations and conversational chat coaching

* Tier 3 - Deterministic Rules Engine: Instant threshold-based notifications for critical stress, low sleep, and dehydration

* Graceful Degradation: If LLM API fails, Tier 1 ML and Tier 3 Rules continue operating seamlessly without app downtime

Slide 4: Machine Learning & Burnout Prediction Pipeline

Timing: 4:00 - 4:45 min

Speaker Script: "Our primary ML model is a Random Forest Classifier trained on 10,000 synthetic developer wellness records incorporating 9 key behavioral features."

Slide Visual Highlights:

* Input Features (9): sleep_hours, working_hours, mood_score, stress_level, energy_level, water_intake, daily_steps, exercise_minutes, consecutive_working_days

* Data Preprocessing: RobustScaler normalizes features using median and IQR, resisting extreme outliers like 16-hour coding days

* Hyperparameter Tuning: GridSearchCV evaluated 18 configurations across 3-fold cross-validation, optimizing macro F1-score

* Output Schema: Risk Class (Low/Medium/High), Confidence %, and Continuous Risk Score (0-100 derived from weighted class probabilities)

Slide 5: Explainable AI (SHAP TreeExplainer)

Timing: 4:45 - 5:30 min

Speaker Script: "A key requirement for user trust in healthcare/wellness AI is explainability. We integrated SHAP (SHapley Additive exPlanations) to make predictions transparent."

Slide Visual Highlights:

* Game Theory Foundation: Calculates exact Shapley value contributions for each feature per prediction

* TreeExplainer: Optimized for tree-based models, computes feature contributions in real-time during API inference

* Human-Readable Factor Translation: Top positive risk contributors are translated to actionable user insights (e.g. 'Low sleep duration', 'Elevated stress index')

* User Empowerment: Developers can see exactly which habits are pushing them into high-risk zones

Slide 6: Generative AI, Prompt Security & Rules Engine

Timing: 5:30 - 7:00 min

Speaker Script: "For creative habit generation and interactive coaching, we integrated Groq's high-speed inference API running LLaMA 3.3-70B."

Slide Visual Highlights:

* Structured JSON Output: Enforces strict response formats using system instructions and response_format parameters

* Developer-Centric Persona: Prompts instruct the LLM to use developer analogies (e.g. 'stack overflow', 'memory leak', 'compile errors')

* AI Security Layer (AISecurityManager): Strips HTML tags, filters 7 prompt injection regex patterns (e.g. 'ignore all prior rules'), and auto-appends medical disclaimers

* Rule-Based Notification Engine: Evaluates 6 threshold rules (e.g. Stress >= 8 and Sleep < 6h triggers 'Nervous System Reset')

Slide 7: Cross-Platform Client Architecture (Flutter)

Timing: 7:00 - 7:45 min

Speaker Script: "The frontend is built using Flutter and Dart, following a clean 3-layer architecture (Data, Domain, Presentation)."

Slide Visual Highlights:

* State Management: Flutter Riverpod for reactive state management and dependency injection

* Declarative Routing: GoRouter with auth-based guards enforcing seamless splash/login/dashboard redirects

* Offline Resilience: Hive key-value database caches metrics locally when offline

* CrashGuard Protection: 3-tier error catching prevents app process crashes even during unexpected platform exceptions

Slide 8: Live Demonstration Setup & Overview

Timing: 7:45 - 11:30 min

Speaker Script: "Now, I will conduct a live demonstration of the application running end-to-end."

Slide Visual Highlights:

* (Proceed to Section 3 of this document for click-by-click live demo script)

* Demonstrate: Auth Login -> Dashboard (11 Cards) -> Log Wellness Entry -> View ML Burnout Prediction & SHAP Factors -> AI Recommendations -> AI Chat Coach -> Telemetry Charts & PDF Export

Slide 9: Security, Cloud Infrastructure & DevOps

Timing: 11:30 - 13:00 min

Speaker Script: "The system is designed with production-grade security and cloud infrastructure."

Slide Visual Highlights:

* Authentication & Security: Firebase Auth issuing JWT ID Tokens, verified backend-side. OWASP security headers (X-Frame-Options: DENY, HSTS, CSP)

* Rate Limiting: Sliding-window rate limiter (60 req/min/IP) preventing API abuse

* Docker Containerization: Single container spec using python:3.12-slim with native HTTP healthchecks

* Cloud Hosting: Render cloud platform running Docker backend service connected to Firebase Firestore

Slide 10: Testing & Quality Assurance

Timing: 13:00 - 14:00 min

Speaker Script: "The backend features a comprehensive test suite written in Pytest."

Slide Visual Highlights:

* Test Coverage: 7 test files covering 25+ test cases

* Validated Areas: Health endpoints, security injection filtering, ML prediction outputs, recommendation feedback, analytics calculations, notification evaluation, PDF generation

* Model Performance: Trained Random Forest model achieves 93% accuracy and 0.93 macro F1-score

Slide 11: Conclusion & Q&A Defense

Timing: 14:00 - 15:00 min

Speaker Script: "In conclusion, MindSync AI demonstrates how combining Machine Learning prediction, Explainable AI, and Generative LLMs can create a tangible solution for software engineer wellness."

Slide Visual Highlights:

* Key Takeaways: 3-Tier Hybrid AI, Real-time SHAP Explainability, Developer-tailored UX, Production Security & Cloud Deployment

* Thank you for your time. I am now open to questions from the panel.

SECTION 3: STEP-BY-STEP LIVE DEMO EXECUTION SCRIPT

Follow this exact step-by-step procedure during the live demo segment of your presentation.

Step 0: How to Launch the Application

Choice A (Full Stack Mode): Run backend server ('python app/main.py' in backend/) and start Flutter client ('flutter run' in frontend/). Choice B (Instant Demo Mode): Set 'static const bool demoMode = true;' in AppConfig.dart to run instant zero-config presentation without backend dependencies.

* Ensure phone/emulator screen is mirrored to the presentation display

* Keep backend terminal visible if showing API logs to technical examiners

Step 1: User Authentication & Splash Screen

Open the app. Point out the splash screen animation and auto-authentication check. Log in using demo credentials or tap 'Sign In as Guest/Demo User'.

* Highlight Firebase Auth integration

* Mention JWT bearer token security

Step 2: Dashboard Exploration

Show the main home screen. Highlight the 11 modular widget cards:

* Welcome Header with user greeting

* Wellness Score Card (0-100 score)

* Burnout Risk Indicator Card

* Mood Summary Card

* Activity & Hydration Summary

* AI Recommendations Carousel

* Weather Info Widget

* Developer Motivational Quote

* Weekly Wellness Charts

* Recent Logs

* Quick Actions Grid

Step 3: Logging a High-Stress Wellness Check-In

Tap the '+ Log Mood' button to open the entry page. Input the following test values specifically engineered to trigger High Burnout Risk alerts:

* Mood: Select Tired/Stressed emoji (Score 3/10)

* Stress Level: Set slider to 9/10

* Energy Level: Set slider to 3/10

* Sleep Hours: Enter 4.5 hours

* Working Hours: Enter 12.0 hours

* Water Intake: 3 glasses (750 ml)

* Daily Steps: 2,000 steps

* Exercise: 0 minutes

* Consecutive Work Days: 6 days

* Tap 'Submit Check-In'

Step 4: Demonstrating ML Prediction & SHAP Explainability

Show the resulting Burnout Prediction Screen:

* Point to Risk Level: 'HIGH' (Red badge)

* Point to Continuous Risk Score: ~82/100

* Point to Confidence: ~91%

* **CRITICAL POINT FOR EXAMINERS:** Highlight the 'Top Contributing Factors' section generated by SHAP TreeExplainer: 'Low sleep duration', 'Elevated stress index', 'Prolonged working hours'

Step 5: Demonstrating Hybrid AI Recommendations

Navigate to the Recommendations tab. Show how recommendations come from 3 distinct sources:

* Rule Engine Rec: 'Nervous System Reset (4-7-8 Breathing)' (triggered by stress > 7)

* ML Rec: 'Mandatory Disconnection' (triggered by High Burnout Risk)

* Groq AI Rec: Creative suggestions generated by LLaMA 3.3-70B

* Demonstrate tapping 'Mark as Completed' or submitting feedback

Step 6: Demonstrating AI Chat Coach

Navigate to the AI Chat tab. Type a message: "I have been debugging a memory leak for 6 hours and feel completely exhausted."

* Show the instant response from MindSync AI

* Point out the developer analogy in the response (e.g. referencing garbage collection or code refactoring)

* Show suggested prompt chips at the bottom

Step 7: Telemetry Analytics & PDF Report Export

Navigate to the Reports tab. Show the 8 interactive fl_chart graphs (Mood, Stress, Sleep, Hydration, Exercise, Burnout, Goals).

* Select 'Last 7 Days' filter

* Tap 'Generate Wellness Report'

* Tap 'Export PDF'

* Show the styled ReportLab PDF opened on screen with telemetry matrices and AI summary

SECTION 4: VIVA EXAMINER QUICK DEFENSE CHEATSHEET

Keep these key technical facts in mind to confidently answer rapid-fire questions during your viva defense.

1. Why Random Forest?

Handles non-linear wellness relationships, resists overfitting via ensemble bagging, outputs probabilities for confidence scoring, and works with SHAP TreeExplainer for real-time explanations.

2. Why RobustScaler?

Uses median and IQR (interquartile range) instead of mean/std dev, preventing extreme outliers (e.g. 16-hour workdays) from skewing feature scaling.

3. Why Synthetic Data?

Real mental health data is restricted under HIPAA/GDPR. Synthetic data generated with NumPy mimics realistic statistical distributions without privacy violations.

4. What is SHAP?

Shapley Additive exPlanations. Game-theoretic metric computing exact feature contributions per prediction. TreeExplainer provides local (per-user) explainability.

5. How does Groq LLM work?

High-speed inference API running Meta's LLaMA 3.3-70B model with structured JSON schema enforcement and developer-centric system instructions.

6. What is the 3-Tier AI System?

Tier 1: Random Forest ML (predicts risk), Tier 2: Groq LLM (generates recommendations/chat), Tier 3: Rules Engine (threshold notifications). Guarantees zero app downtime.

7. What security features are built-in?

Firebase JWT auth token verification, AISecurityManager (HTML stripping, 7 prompt injection regex filters), OWASP security headers, and sliding-window rate limiting (60 req/min/IP).

8. How does the Wellness Score work?

Daily Score = 30% Mood + 20% Inverse Stress + 20% Sleep (cap 8h) + 15% Hydration (cap 8 glasses) + 15% Exercise (cap 30m). Range: 0-100.

9. What is Clean Architecture?

3-layer design pattern in Flutter: Data (API/models) -> Domain (entities/use cases) -> Presentation (UI/Riverpod providers). Dependencies point inward.

10. What emergency backup exists if API fails during Viva?

AppConfig.dart has a boolean 'demoMode = true' flag that instantly switches the entire app to local offline mock data without requiring internet or backend servers.